

Multimodal Fashion & Context Retrieval

Natural-language fashion image search across garments, color, style, and scene context

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GitHub repository	github.com/aayushi262004/Glance_internship_assignment
Demo video	loom.com/share/8fa686c16dc34c77bb977883578b028f
Corpus	1,000 Fashionpedia images · 4,027 annotated garment regions · 27 categories
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Headline result. A decomposed, region-grounded retrieval architecture with a Marqo-FashionSigLIP backbone and pixel-level color verification reaches **Precision@5 0.960** and **mAP 0.991** on the 15-query human-judged evaluation (vanilla CLIP baseline: 0.867 / 0.964), and scores a **perfect P@5 = 1.0 on four of the five mandatory assignment queries**. Every design decision in this report is backed by a measured A/B under an identical human-labeling protocol — including two experiments that were *rejected* because the numbers said so.

1 Problem & Design Goal

The task is a search engine over a fashion corpus answering free-text queries that combine *what* is worn (garments, colors), the *vibe* (style), and *where* it is worn (scene context) — e.g. “a bright yellow raincoat”, “a red tie and a white shirt in a formal setting”, “someone wearing a blue shirt sitting on a park bench”.

The central ML problem: global CLIP is a strong zero-shot baseline — my own measured vanilla baseline reaches Precision@5 = 0.867 on this corpus — but it compresses the whole image and the whole query into single vectors. Two failure modes follow directly: **(i) dilution** — a tie occupying 2% of the frame barely moves the global embedding; and **(ii) loose attribute binding** — a query mentioning “red” and “tie” can be satisfied in embedding space by a red outfit with no tie, because nothing forces the color to bind to the garment. The assignment hint names exactly this compositionality gap, and the architecture below is built to close it while keeping CLIP’s broad zero-shot power.

2 Approaches Considered & Trade-offs (Deliverable 1)

A. Vanilla global CLIP retrieval

Encode every image and the full query with CLIP; rank by cosine similarity in one FAISS index. **Good:** simple, strong zero-shot semantics, no training, trivially scalable. **Weak:** one vector per side dilutes small garments; attribute→garment binding is implicit and fails on compositional queries. **When:** the right *first system* and the honest baseline every improvement must beat — I built and measured it (P@5 0.867).

B. Fashion-specialized global encoder (FashionCLIP / FashionSigLIP)

Swap CLIP for a frozen fashion-adapted encoder. **Good:** better garment vocabulary and fine-grained fashion similarity at zero training cost. **Weak:** weaker scene/context semantics than general CLIP; still one global vector, so composition is still unverified. **When:** fashion-dominant corpora — used here as one signal, not the whole system.

C. Detection / segmentation + symbolic attribute classifiers

Detect garments, classify color/type per garment, match queries against structured tags. **Good:** exact, auditable attribute binding. **Weak:** closed vocabulary kills zero-shot capability (“business attire”, “casual weekend vibe” have no tag); pipeline drift across detector + classifiers; misses style/scene nuance. **When:** catalog/e-commerce data with clean product attributes — wrong fit for open natural-language search.

D. Captioning / VLM + LLM reasoning per query

Caption images offline (or run a VLM per query) and let an LLM match text to text. **Good:** strongest relational reasoning (“sitting on a bench”). **Weak:** captions lossy for fine-grained fashion detail; per-query VLM scoring is orders of magnitude more expensive and does not scale to 1M images as a first stage. **When:** as a *reranker* over a small candidate pool — exactly where it appears in my future-work plan (§7).

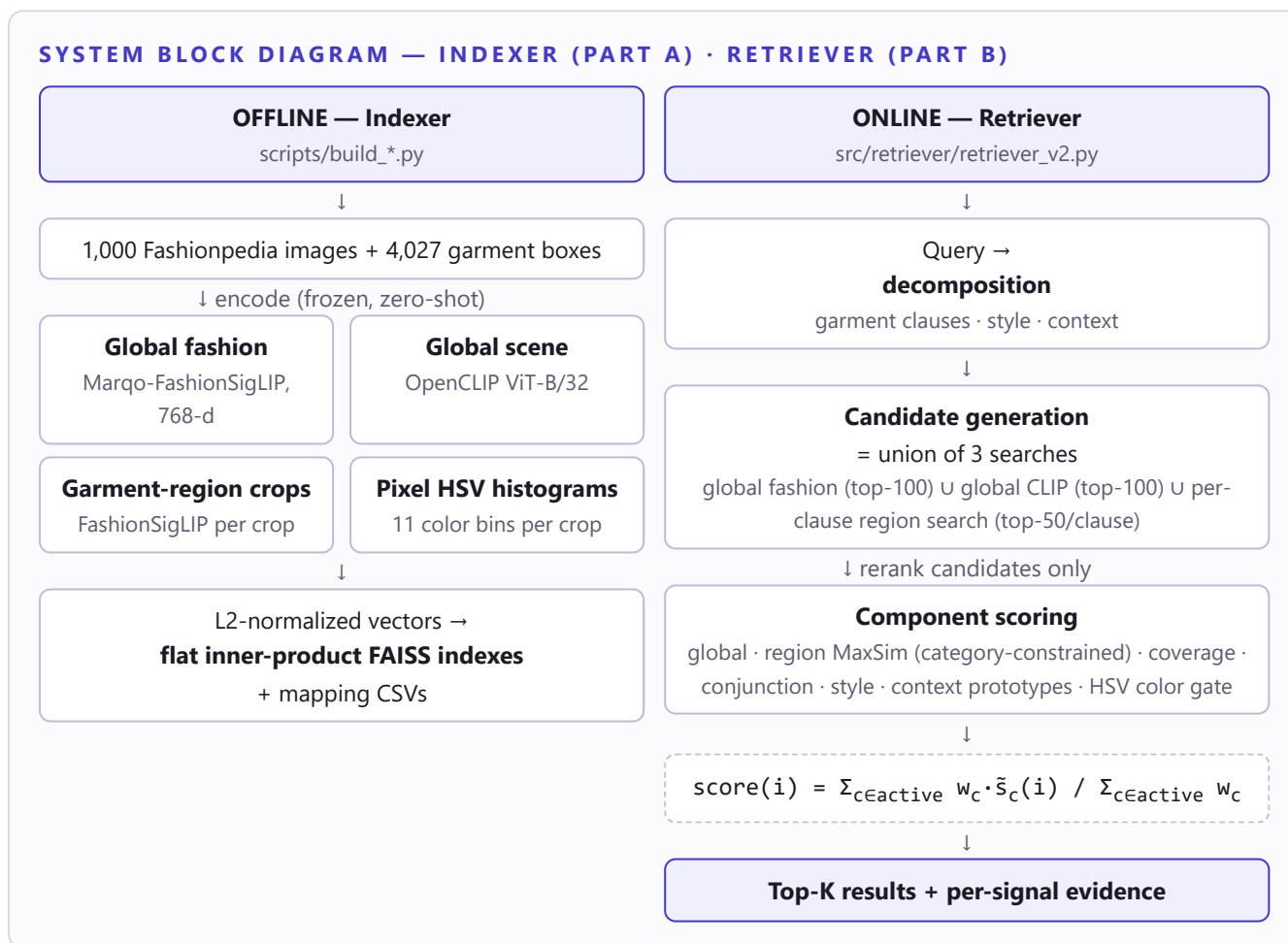
E. Fine-tuning / learned reranker future work

Good: can learn true compositional binding. **Weak:** needs relevance labels at a scale that does not exist here (75 judgments); risks overfitting 1,000 images and losing zero-shot generality. **When:** after production interaction logs accumulate.

F. Decomposed multi-signal retrieval with garment-region grounding chosen

Keep frozen zero-shot encoders but use each where it is strongest, and ground garment clauses in *region-level* evidence: a fashion encoder for garment semantics, general CLIP for scene context, fashion-encoder embeddings of every annotated garment crop as explicit per-garment evidence, and pixel-HSV histograms for color verification — combined by query-adaptive late fusion. **Good:** verifies each clause against the right image region (fixes dilution + binding), retains zero-shot behavior as its fallback, fully interpretable (per-component scores per result), everything stays FAISS-indexable. **Weak:** needs region annotations (or a detector) and a query parser; more moving parts than A/B. **When:** compositional, attribute-heavy fashion search — this assignment’s core.

3 Chosen Architecture (Deliverable 2)



3.1 Query decomposition

A lightweight rule-based parser splits the query into **garment clauses** mapped onto the Fashionpedia taxonomy (*"red tie"* → category *tie*; *"white shirt"* → *shirt/blouse, top*), a **style** phrase (*formal, casual, business...*) and a **context** phrase (*office, park, street, home...*). Unknown vocabulary falls through to whole-query zero-shot retrieval — decomposition strengthens what it recognizes and never breaks what it doesn't.

3.2 Candidate generation, then reranking

Candidates are the **union** of global fashion search, global CLIP search, and — per clause — a search of the garment-region index. The region path is the recall backbone for composition: an image whose global

embedding ranks poorly still enters the pool because one crop of it matches “red tie”. All subsequent scoring touches only this pool (a few hundred images), which is what makes the logic scale-independent (§8).

3.3 Region evidence: category-constrained MaxSim

Per candidate, each clause is scored only against that image’s crops of *compatible* categories (a tie clause against tie crops), taking the maximum similarity. This is the explicit attribute-to-garment binding vanilla CLIP lacks.

3.4 Coverage, conjunction, and adaptive calibration

Multi-garment queries reward satisfying *all* constraints: a coverage score counts matched clauses; a conjunction bonus fires only when every clause matches. Whether a clause “matches” is decided against the **percentile of that clause’s own score distribution across the current candidate pool** (plus an absolute floor) rather than a universal cosine threshold — rare categories produce systematically different score ranges than common ones, so a fixed threshold either starves rare garments or floods common ones.

3.5 Style and contrastive context prototypes

Style is scored with the fashion encoder against the whole image. Context is scored with general CLIP against a **contrastive prototype bank**: mean similarity to positive prompts (“*professional office interior*”) minus a weighted maximum over negatives (“*outdoor street*”, “*fashion runway*”) — a sharper scene signal than any single embedding, and the template that later extends to cities and weather (§6).

3.6 Pixel-HSV color verification (the color-binding fix)

CLIP-family crop similarity is shape-dominated: a red jacket scores nearly as high as blue ones for “blue jacket”. Instead of fighting embeddings with embeddings, color is verified from **actual crop pixels**: each region carries a precomputed 11-bin HSV histogram, and color clauses apply a soft multiplicative gate on region evidence. The key subtlety is a **perceptual color-adjacency map** — golden/mustard hues are credited as yellow-adjacent, navy as blue-adjacent — so the gate catches clear mismatches (red≠blue, demoting a human-confirmed false positive from rank #2) without the yellow/beige regression that sank an earlier embedding-based color experiment (§4.3).

3.7 Query-adaptive late fusion

Only the components the query actually contains are activated, min-max normalized over the candidate pool, and combined with weights renormalized over active components (global 0.25 · region 0.30 · coverage 0.15 · conjunction 0.10 · style 0.15 · context 0.15). A garment-free query like “*casual weekend outfit for a city walk*” gracefully reduces to global + style + context retrieval.

3.8 How each mandatory query type is handled

Assignment query type	Mechanism that answers it
Attribute — “bright yellow raincoat”	Region MaxSim on <i>coat</i> crops + HSV gate verifying yellow-family pixels
Contextual — “business attire in a modern office”	Style component + contrastive office-vs-street/runway prototypes on general CLIP
Complex semantic — “blue shirt on a park bench”	Region evidence for <i>blue shirt</i> fused with park-context prototypes
Style inference — “casual weekend outfit for a city walk”	No garment clause → fusion reduces to global + style + context (zero-shot path)
Compositional — “red tie AND white shirt, formal”	Per-clause region search + category-constrained MaxSim + coverage/conjunction reward for satisfying <i>both</i> clauses

4 Measured Results & Evidence-Driven Model Selection

15 queries (the five mandatory prompts + ten garment/compositional/style/context queries), top-5 per query, graded human relevance labels (0/1/2) under one rubric, maintained in a reusable label store (`evaluation/harness/`) that **refuses to compute metrics over unlabeled pairs**. Metrics: Precision@5 (rel $\geq$ 1), mAP bounded to the top-5 pool, nDCG@5 on graded labels.

4.1 System progression

Configuration	Precision@5	mAP@5	nDCG@5
Vanilla global CLIP (whole query $\leftrightarrow$ whole image)	0.8667	0.9644	0.9779
Decomposed + region fusion (FashionCLIP backbone)	0.9333	0.9631	0.9627
+ pixel-HSV color gate	0.9467	0.9719	0.9680
Shipped: Marqo-FashionSigLIP backbone + color gate	0.9600	0.9911	0.9618

The architecture (row 2) lifts P@5 by +6.7 points over vanilla CLIP; the backbone swap to Marqo-FashionSigLIP (a stronger frozen fashion-adapted SigLIP, richer 768-d embeddings, identical zero-shot usage) adds the largest single gain, moving **both** P@5 and mAP together — the trustworthy signal at this evaluation size. On “*yellow raincoat*” the vanilla baseline scores nDCG = 1.0 while retrieving only two relevant images in five — pool-bounded nDCG rewards ordering of whatever was retrieved; Precision@5 is the discriminative metric here, and it is where the architecture shows.

4.2 The five mandatory assignment queries (P@5)

Query	Type	Vanilla CLIP	Shipped system
“A person in a bright yellow raincoat”	Attribute	0.4	0.8
“Professional business attire inside a modern office”	Contextual	1.0	1.0
“Someone wearing a blue shirt sitting on a park bench”	Complex semantic	0.8	1.0
“Casual weekend outfit for a city walk”	Style inference	1.0	1.0
“A red tie and a white shirt in a formal setting”	Compositional	0.6	1.0

The largest gains land exactly on the two failure modes the architecture targets: the fine-grained attribute query (0.4 $\rightarrow$ 0.8, doubling relevant results) and the compositional query (0.6 $\rightarrow$ **1.0**, perfect despite only two annotated ties existing in the entire corpus — the region path finds both). The remaining raincoat gap is warm-tone (yellow/beige/camel) confusion at the pixel-hue boundary, stated honestly in §9.

4.3 Measured and rejected — model selection, not intuition

- **Contrastive CLIP color prompts** (discount a “blue jacket” crop by its similarity to other colors): improved compositional nDCG 0.912 $\rightarrow$ 0.945 but cut overall P@5 0.933 $\rightarrow$ 0.907 and regressed yellow/beige queries. **Rejected**; replaced by the pixel-HSV gate, which fixes the same failure without the regression.
- **CLIP-global as a scoring component** (baseline uses it for recall and context only): improved mAP but cut P@5 to 0.920 by diluting fine-grained garment queries. **Rejected**.

- Both experiments' labeled artifacts live in the repo (`evaluation/experiments/` , `evaluation/harness/`) for reproduction.

5 Codebase (Deliverable 3)

GitHub: https://github.com/aayushi262004/Glance_internship_assignment

Logic is separated from data, and indexing from retrieval, per the assignment's modularity ask:

Module	Responsibility
<code>scripts/</code>	Part A — Indexer. Dataset preparation; builds global CLIP / FashionCLIP / Marqo indexes, garment-region index, HSV color histograms
<code>src/indexer/</code>	Frozen encoder wrappers (OpenCLIP, FashionCLIP)
<code>src/retriever/</code>	Part B — Retriever. Query decomposition, candidate generation, region scoring, calibration, color gate, late fusion (<code>retriever_v2.py</code>)
<code>evaluation/</code> + <code>evaluation/harness/</code>	Queries, human label store, metrics, reproducible per-config runs, rejected-experiment artifacts
<code>server/</code> + <code>web/</code>	FastAPI API + React demo UI: parsed-query view, per-result component score bars, live backbone & color-gate ablation toggles, fusion-weight sliders, image-to-image similar-looks

Reproduce: `pip install -r requirements.txt` → run the five `scripts/build_*.py` builders → `./run.sh` (starts API + UI). Exact tested versions in `requirements-lock.txt` ; evaluation via `evaluation/harness/run_config.py` .

6 Future Work A — Locations (Cities, Places) & Weather (Deliverable 4a)

The late-fusion design makes these *additive* extensions: each new axis is one more query-conditioned component plus (when tags exist) one metadata filter — no re-architecting, and existing signals are untouched.

EXTENSION PATTERN — EVERY NEW AXIS IS A FUSION COMPONENT + OPTIONAL METADATA FILTER

Query: "someone in a trench coat on a
rainy London
street"

↓ parse (garment || style || context || **location** || **weather**)

Existing signals

garment regions · style · scene

Location component

place-recognition embeddings
(landmark / Places-style) + per-city
contrastive prototypes

Weather component

rain/snow/sun/fog prototypes or
lightweight weather classifier

↓ hard filter first when tags exist: geo/EXIF → city, timestamp × weather-history join

Same late fusion, weights renormalized over active components → Top-K

Locations

- **Visual path (no metadata needed):** add a place-recognition embedding signal — a Places-trained scene classifier or landmark retrieval embeddings — as a component activated only for location-bearing queries, with per-city/place contrastive prototype banks (positives: "Tokyo shopping street at night"; negatives: "generic mall interior"). This mirrors exactly how the office/park context component already works, so wiring cost is minimal.
- **Metadata path (when available):** geotag/EXIF → reverse-geocoded city becomes a hard pre-filter before vector search (city shard or filtered index) — cheap, exact, and it shrinks the candidate pool rather than growing the scoring problem.
- Named-place vocabulary enters the parser the same way garments did; unknown places fall back to whole-query zero-shot, preserving graceful degradation.

Weather

- Weather concepts (rainy, snowy, sunny, foggy) fit the identical pattern: contrastive visual-text prototypes or a small frozen weather classifier as a query-conditioned component.
- Weather also **pairs with garment evidence**: raincoats and umbrellas are already Fashionpedia categories, so "rainy" queries get corroborating region signal for free — an interaction unique to this decomposed design.
- With timestamps + location, a weather-history join gives ground-truth tags for hard filtering; the visual component covers untagged corpora.

7 Future Work B — Improving Precision (Deliverable 4b)

Ordered by expected impact per unit cost, given where measured errors actually live:

1. **VLM / cross-encoder reranking of the top ~50 candidates.** mAP is already 0.991, so remaining headroom is in the hardest tail: relational queries and near-duplicate distinctions. A vision-language model scoring only ~50 images per query keeps cost bounded and independent of corpus size, and directly verifies relations ("*sitting on a bench*") that embeddings only correlate with.
2. **Explicit relation/action verification.** A pose estimator or human-object-interaction model as a boolean/soft component for action clauses (sitting, walking, holding) — the park-bench query is currently answered by correlation, not verification.
3. **Open-vocabulary garment grounding.** Map unseen garment words onto the Fashionpedia taxonomy by embedding similarity ("*windcheater*" → jacket/coat), extending region grounding beyond the parser's closed vocabulary; longer term, replace annotation dependence with an open-vocabulary detector (e.g. SAM/OWL-style) so region evidence exists for unannotated corpora.
4. **Learned fusion and held-out calibration.** Replace hand-set weights with weights fit on accumulated relevance judgments; calibrate the clause-match percentile on a held-out split.
5. **Scale the evaluation itself.** More queries and multiple labelers — at 15 queries a single image is ± 0.067 P@5, so a larger multi-labeler set is what makes further deltas statistically meaningful (precision improvements must be *measurable* to be real).
6. **Corpus coverage.** The office and tie sparsity are data problems no ranking change fully fixes; targeted corpus expansion lifts the ceiling for every component at once.

8 Scalability to 1M Images & Zero-Shot Capability

Scalability. Flat exact FAISS indexes are the right call at 1,000 images (the assignment de-prioritizes index engineering). At ~1M the *retrieval logic is unchanged* and only the index engineering swaps: IVF/HNSW (optionally PQ-compressed) for the global indexes; garment-region vectors sharded by category so a tie clause searches only the tie shard; distributed offline encoding; metadata in a real store. Crucially, all reranking (region MaxSim, calibration, fusion) is already bounded to the candidate pool — query cost grows with candidates, not corpus size — so the two-stage shape needs no redesign.

Zero-shot capability. Every encoder is frozen; nothing is trained on this corpus. The parser strengthens recognized vocabulary; anything outside it — unseen garments, styles, places — still retrieves through whole-query embeddings. The system therefore *degrades to a strong zero-shot baseline instead of failing*, which is the property that lets the same design absorb new axes (locations, weather) without retraining.

9 How This Is Useful for Glance

Glance's surfaces are content-first: fashion and lifestyle imagery meets users in feeds and on the lock screen, where discovery is visual and intent arrives as natural language, not catalog filters. This system is built for exactly that setting:

- **Natural-language fashion discovery.** Users (or editorial tooling) can query looks the way people actually talk — "*office-ready blazer look*", "*casual weekend outfit*" — and get garment-verified results, not just vibe-similar images. The same engine powers user-facing search, editorial curation, and content tagging pipelines.
- **Zero-shot onboarding of new content.** Every encoder is frozen: new creators, drops, or catalog imagery become searchable the moment they are indexed — no retraining cycle stands between content ingestion and discovery, which matters at feed-scale content velocity.

- **Shoppable-content and recommendation hooks.** The garment-region index is image-to-garment structure: the same crops that verify “red tie” can anchor product matching, similar-looks recommendations (already live in the demo UI), and outfit-level content clustering for feed ranking.
- **Contextual by design.** The location/weather extension (§6) maps directly onto Glance’s context-aware surface: a lock screen that knows it is a rainy morning in Mumbai can retrieve monsoon-ready looks with one added fusion component — the architecture is already shaped for query-conditioned context signals.
- **Interpretable ranking for a content business.** Every result carries per-component evidence (which garment matched, which crop, what context score). Merchandising and trust teams can see *why* an image ranked — a practical requirement the moment retrieval decisions touch revenue surfaces.
- **A serving shape that survives growth.** The two-stage candidate-union + bounded-rerank design (§8) is the same shape production feed retrieval uses, so the path from this prototype to a Glance-scale service is index engineering, not a redesign.

10 Demo Video

An end-to-end walkthrough of the system — query input, parsed garment/style/context interpretation, top-5 retrieval for the five mandatory assignment queries, per-component score evidence, live backbone and color-gate ablation toggles, and image-to-image similar looks:

Demo video: <https://www.loom.com/share/8fa686c16dc34c77bb977883578b028f>

11 Limitations, Stated Honestly

- **Warm-tone color boundary.** Yellow/beige/camel confusion persists at the hue boundary (raincoat P@5 0.8, not 1.0); pixel hue separates clear mismatches, not perceptual near-neighbors.
- **Relations are inferred, not verified.** No pose/spatial model yet (§7.2).
- **Annotation dependence.** Region evidence exists only where garments are annotated; unannotated garments contribute global signal only.
- **Evaluation scale.** 15 queries / 75 judgments, largely one labeler; per-query deltas of one image are within labeling noise — which is why decisions favored movements confirmed by multiple metrics.
- **Hand-set fusion weights and calibration percentile** (not learned; §7.4).
- **Corpus coverage.** Two annotated ties and thin office imagery bound every component.

Conclusion. Vanilla CLIP is a strong baseline — measurably so — but decomposing fashion queries and grounding every garment clause in category-aware region evidence, verified by actual pixels and fused adaptively, improves top-k relevance exactly where global embeddings are weakest. The shipped system reaches P@5 0.960 / mAP 0.991 with perfect top-5 precision on four of five mandatory queries, every claim carries a measured number, and two tempting ideas were rejected because the evaluation said so. That — an architecture chosen by evidence, with its remaining failures named — is the ML logic this report set out to demonstrate.